C2.3 Additional response signals and signaling requirements defined by ACE

The following subsections introduce the additional response signals, and an additional signalling requirement, introduced by ACE:

- *Read acknowledge signal*
- *Write acknowledge signal*
- *Additional reset requirements*.

C2.3.1 Read acknowledge signal

Table C2-7 shows the additional read acknowledge signal. See *Read acknowledge signaling* on page C3-170.

Table C2-7 Read acknowledge signal

Signal	Source	Description
RACK	Master	Read acknowledge. This signal indicates that a master has completed a read transaction.

C2.3.2 Write acknowledge signal

Table C2-8 shows the additional write acknowledge signal. See *Write Acknowledge signaling* on page C3-172.

Table C2-8 Write acknowledge signal

Signal	Source	Description
WACK	Master	Write acknowledge. This signal indicates that a master has completed a write transaction.

C2.3.3 Additional reset requirements

The ACE protocol uses the AXI4 single active LOW reset signal **ARESETn**. See *Reset* on page A3-36 for the **ARESETn** requirements.

During reset, the following interface requirements apply:

- a master interface must drive **RACK**, **WACK**, **CRVALID**, and **CDVALID** LOW
- an interconnect must drive **ACVALID** LOW.

The earliest point after reset that the interconnect is permitted to begin driving **ACVALID** HIGH is at a rising **ACLK** edge after **ARESETn** is HIGH.

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Chapter C3
Channel Signaling

This chapter describes the basic channel signaling requirements on an ACE interface. It contains the following sections:

- *Read and write address channel signaling* on page C3-158
- *Read data channel signaling* on page C3-167
- *Read acknowledge signaling* on page C3-170
- *Write response channel signaling* on page C3-171
- *Write Acknowledge signaling* on page C3-172
- *Snoop address channel signaling* on page C3-173
- *Snoop response channel signaling* on page C3-176
- *Snoop data channel signaling* on page C3-180
- *Snoop channel dependencies* on page C3-182.

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